

Лабораторная работа 7

КОНТРОЛЬ ИСПОЛЬЗОВАНИЯ РЕСУРСОВ ОС LINUX

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1. Вывести список всех процессов системы.

```
ubuntu@ubuntu:~$ ps aux
USER          PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root             1  0.8  0.4 23388  8592 ?        Ss   12:00   0:06 /sbin/init au
root             2  0.0  0.0      0     0 ?        S    12:00   0:00 [kthreadd]
root             3  0.0  0.0      0     0 ?        S    12:00   0:00 [pool_workque
root             4  0.0  0.0      0     0 ?        I<   12:00   0:00 [kworker/R-rc
root             5  0.0  0.0      0     0 ?        I<   12:00   0:00 [kworker/R-sy
root             6  0.0  0.0      0     0 ?        I<   12:00   0:00 [kworker/R-kv
root             7  0.0  0.0      0     0 ?        I<   12:00   0:00 [kworker/R-sl
root             8  0.0  0.0      0     0 ?        I<   12:00   0:00 [kworker/R-ne
root            10  4.5  0.0      0     0 ?        I    12:00   0:34 [kworker/0:1-
root            11  0.0  0.0      0     0 ?        I<   12:00   0:00 [kworker/0:0H
root            12  1.5  0.0      0     0 ?        I    12:00   0:11 [kworker/u4:0
root            13  0.0  0.0      0     0 ?        I<   12:00   0:00 [kworker/R-mm
root            14  0.0  0.0      0     0 ?        I    12:00   0:00 [rcu_tasks_kt
root            15  0.0  0.0      0     0 ?        I    12:00   0:00 [rcu_tasks_ru
root            16  0.0  0.0      0     0 ?        I    12:00   0:00 [rcu_tasks_tr
root            17  1.8  0.0      0     0 ?        S    12:00   0:13 [ksoftirqd/0]
root            18  0.4  0.0      0     0 ?        I    12:00   0:03 [rcu_preempt]
root            19  0.0  0.0      0     0 ?        S    12:00   0:00 [rcu_exp_par_
root            20  0.0  0.0      0     0 ?        S    12:00   0:00 [rcu_exp_gp_k
root            21  0.0  0.0      0     0 ?        S    12:00   0:00 [migration/0]
root            22  0.0  0.0      0     0 ?        S    12:00   0:00 [idle_inject/
root            23  0.0  0.0      0     0 ?        S    12:00   0:00 [cpuhp/0]
```

2. Вывести дерево процессов.

```
ubuntu 6298 0.0 0.0 452 136 ?        D    12:13   0:00 /snap/snap-st
ubuntu@ubuntu:~$ pstree
systemd--ModemManager--3*[{ModemManager}]
--NetworkManager--3*[{NetworkManager}]
--accounts-daemon--3*[{accounts-daemon}]
--avahi-daemon--avahi-daemon
--casper-md5check
--colord--3*[{colord}]
--cron
--cups-browsed--3*[{cups-browsed}]
--cupsd
--dbus-daemon
--gdm3--gdm-session-wor--gdm-x-session--Xorg--{Xorg}
--gnome-session-b--3*[{gnome-session-b}]
--3*[{gdm-x-session}]
--3*[{gdm-session-wor}]
--3*[{gdm3}]
--gnome-remote-de--3*[{gnome-remote-de}]
--2*[kerneloops]
--polkitd--3*[{polkitd}]
--power-profiles--3*[{power-profiles-}]
--python3.10--python3.10--rsync--rsync--rsync
--rsyslogd--3*[{rsyslogd}]
--rtkit-daemon--2*[{rtkit-daemon}]
```

3. С помощью команды top получить список 5 процессов, потребляющих наибольшее количество процессорного времени.

Командой top вывожу динамически обновляющийся список всех процессов

```
ubuntu@ubuntu:~$ top

top - 12:32:12 up 31 min, 1 user, load average: 3.70, 4.34, 5.33
Tasks: 217 total, 2 running, 215 sleeping, 0 stopped, 0 zombie
%Cpu(s): 3.7 us, 7.0 sy, 0.9 ni, 0.0 id, 74.3 wa, 0.0 hi, 14.0 si, 0.0 st
MiB Mem : 1968.7 total, 86.4 free, 1886.4 used, 775.7 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 82.4 avail Mem

  PID USER      PR  NI  VIRT  RES  SHR  S  %CPU  %MEM    TIME+  COMMAND
 6897 root        20   0 342816 129352 13156 D   26.5   6.4   0:03.74 appstreamcli
 2516 ubuntu     20   0 3454040 239556 23780 S    3.0  11.9   1:27.75 gnome-shell
 4118 root        39  19   3776   904   648 D    2.3   0.0   0:42.34 casper-md5check
  47 root        20   0     0     0     0 S    1.7   0.0   0:15.37 kswapd0
 6387 root        20   0     0     0     0 D    1.3   0.0   0:00.15 kworker/u4:3+flush-+
 6418 root         0 -20     0     0     0 I    1.3   0.0   0:05.78 kworker/0:0H-kblockd
 6424 root        20   0     0     0     0 I    1.3   0.0   0:02.63 kworker/0:0-ata_sff
  12 root        20   0     0     0     0 I    0.3   0.0   0:22.27 kworker/u4:0-ext4-r+
  17 root        20   0     0     0     0 S    0.3   0.0   0:32.26 ksoftirqd/0
1278 root        39  19     0     0     0 S    0.3   0.0   0:00.19 arc_reap
1315 root        20   0     0     0     0 I    0.3   0.0   0:18.37 kworker/u4:8-events+
```

Нажимаю f для сортировки, после чего выбирают time+, затем q для выхода из меню. Теперь список отсортирован по потреблению времени.

```
Fields Management for window 1:Def, whose current sort field is TIME+
Navigate with Up/Dn, Right selects for move then <Enter> or Left commits,
'd' or <Space> toggles display, 's' sets sort. Use 'q' or <Esc> to end!

* PID      = Process Id   PGRP      = Process Gr   OOMs        = OOMEM Scor  RSS          = Res Mem (s
* USER     = Effective   TTY        = Controllin  ENVIRON      = Environmen  PSS          = Proportion
* PR        = Priority    TPGID      = Tty Proces  vmj          = Major Faul  PSan         = Proportion
* NI        = Nice Value  SID        = Session Id  vMn         = Minor Faul  PSfd         = Proportion
* VIRT      = Virtual Im  nTH        = Number of   USED        = Res+Swap S  PSsh         = Proportion
* RES       = Resident S  P          = Last Used  nsIPC       = IPC namesp  USS          = Unique RSS
* SHR       = Shared Mem  TIME       = CPU Time  nsMNT       = MNT namesp  ioR          = I/O Bytes
* S         = Process St  SWAP       = Swapped Si  nsNET       = NET namesp  ioRop        = I/O Read O
* %CPU      = CPU Usage   CODE       = Code Size  nsPID       = PID namesp  ioW          = I/O Bytes
* %MEM      = Memory Usa  DATA      = Data+Stack nsUSER      = USER names ioWop        = I/O Write
* TIME+     = CPU Time,   nMaj       = Major Page nsUTS       = UTS namesp  AGID         = Autogroup
* COMMAND   = Command Na nMin       = Minor Page  LXC         = LXC contai AGNI         = Autogroup
* PPID      = Parent Pro nDRT       = Dirty Page  RSan        = RES Anonym  STARTED     = Start Time
* UID       = Effective  WCHAN      = Sleeping i  RSfd        = RES File-b ELAPSED     = Elapsed Ru

top - 12:41:36 up 40 min, 1 user, load average: 1.70, 2.22, 3.84
Tasks: 213 total, 1 running, 212 sleeping, 0 stopped, 0 zombie
%Cpu(s): 1.2 us, 22.1 sy, 0.0 ni, 0.0 id, 66.0 wa, 0.0 hi, 10.7 si, 0.0 st
MiB Mem : 1968.7 total, 91.5 free, 1823.2 used, 845.2 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 145.5 avail Mem

  PID USER      PR  NI  VIRT  RES  SHR  S  %CPU  %MEM    TIME+  COMMAND
    1 root        20   0  23228  8348  3484 S   0.0   0.4   6:36.06 systemd
 2516 ubuntu     20   0 3454040 242892 27256 S   1.0  12.0   1:53.69 gnome-shell
 2145 ubuntu     20   0  21568  5336 2008 S   0.0   0.3   1:42.97 systemd
 6625 root        20   0  61072 24076 1592 S   0.0   1.2   1:13.18 python3.10
1654 root        20   0 1847964 15000  420 S   0.0   0.7   1:08.12 snapd
  17 root        20   0     0     0     0 S   0.0   0.0   0:39.46 ksoftirqd/0
2293 ubuntu     20   0  318232 52252 13360 S   1.3   2.6   0:37.35 Xorg
4094 root        20   0  646852 62156 3908 S   0.0   3.1   0:26.86 python3.10
  12 root        20   0     0     0     0 I   0.3   0.0   0:24.34 kworker/u4:0-ext4-r+
6393 root        20   0     0     0     0 I   0.0   0.0   0:23.90 kworker/0:2-cgroup_+
1315 root        20   0     0     0     0 I   0.0   0.0   0:20.35 kworker/u4:8-events+
  47 root        20   0     0     0     0 S   0.3   0.0   0:16.97 kswapd0
7733 root        20   0  43696 28508 2756 S  19.3   1.4   0:14.58 dpkg
6424 root        20   0     0     0     0 I   0.0   0.0   0:14.51 kworker/0:0-cgroup_+
6151 ubuntu     20   0  776548 18744  7560 S   0.3   0.9   0:13.72 gnome-terminal-
```

После нажимаю n и ввожу число 5 чтобы оставить только 5 верхних строчек.

```
top - 12:42:18 up 41 min, 1 user, load average: 1.41, 2.08, 3.72
Tasks: 213 total, 2 running, 211 sleeping, 0 stopped, 0 zombie
%Cpu(s): 1.2 us, 30.5 sy, 0.0 ni, 0.0 id, 56.0 wa, 0.0 hi, 12.4 si, 0.0 st
MiB Mem : 1968.7 total, 83.0 free, 1826.1 used, 851.8 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 142.6 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
2516	ubuntu	20	0	3454040	243352	27716	S	4.3	12.1	1:42.07	gnome-shell
17	root	20	0	0	0	0	S	0.0	0.0	0:39.55	ksoftirqd/0
2293	ubuntu	20	0	318232	52896	14004	S	4.3	2.6	0:37.97	Xorg
12	root	20	0	0	0	0	I	1.0	0.0	0:24.68	kworker/u4:0-writeb+
1654	root	20	0	1847964	18392	3812	S	0.0	0.9	0:24.13	snaped

4. Найти 2 процесса, имеющих более двух потоков. Использовать состояние процесса.

Ввожу команду top, после чего нажимаю o для фильтра и ввожу S=I, чтобы отображались только процессы с двумя и более потоками.

```
ubuntu@ubuntu: ~
top - 12:55:55 up 55 min, 1 user, load average: 1.72, 1.69, 2.64
Tasks: 218 total, 3 running, 212 sleeping, 0 stopped, 3 zombie
%Cpu(s): 3.7 us, 34.6 sy, 0.0 ni, 0.0 id, 0.7 wa, 0.0 hi, 61.0 si, 0.0 st
MiB Mem : 1968.7 total, 77.2 free, 1897.7 used, 771.9 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 71.0 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
7732	root	20	0	0	0	0	I	3.4	0.0	0:10.66	kworker/0:1-ata_sff
18	root	20	0	0	0	0	I	0.6	0.0	0:07.00	rcu_preempt
13784	root	20	0	0	0	0	I	0.3	0.0	0:00.43	kworker/u4:3-events+
4	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-rcu_gp
5	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-sync_wq
6	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-kvfree_rc+
7	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-slub_flush+
8	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-netns
12	root	20	0	0	0	0	I	0.0	0.0	0:25.36	kworker/u4:0-loop2
13	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-mm_percpu+
14	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_kthread
15	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_rude_kthr+
16	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_trace_kth+
25	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-inet_frag+
31	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-writeback
35	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-kintegrity

```
top - 12:58:49 up 58 min, 1 user, load average: 11.89, 5.42, 3.82
Tasks: 223 total, 2 running, 219 sleeping, 0 stopped, 2 zombie
%Cpu(s): 3.2 us, 44.6 sy, 0.0 ni, 0.0 id, 36.9 wa, 0.0 hi, 15.3 si, 0.0 st
MiB Mem : 1968.7 total, 77.6 free, 1883.5 used, 804.9 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 85.2 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
4	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-rcu_gp
5	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-sync_wq

5. Используя команду top, изменить приоритеты 2 процессов.

После вызова команды sudo top, чтобы вызвать команду в режиме суперпользователя, нажимаю r и ввожу pid процесса.

```
top - 12:59:49 up 59 min, 1 user, load average: 6.30, 4.96, 3.76
Tasks: 225 total, 1 running, 222 sleeping, 0 stopped, 2 zombie
%Cpu(s): 0.0 us, 29.4 sy, 5.9 ni, 0.0 id, 47.1 wa, 0.0 hi, 17.6 si, 0.0 st
MiB Mem : 1968.7 total, 75.8 free, 1908.8 used, 759.3 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 59.9 avail Mem
PID to renice [default pid = 14928] 8
  PID USER      PR  NI  VIRT  RES  SHR S %CPU  %MEM    TIME+  COMMAND
 14928 root        39  19 210776 70472 29184 D  4.8   3.5   0:03.45 apt-check
    1 root         20   0  23484   5940  1076 S   0.0   0.3   0:16.30 systemd
    2 root         20   0      0      0      0 S   0.0   0.0   0:00.00 kthreadd
    3 root         20   0      0      0      0 S   0.0   0.0   0:00.00 pool_workqueue_rele+
    4 root          0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/R-rcu_gp
    5 root          0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/R-sync_wq
    6 root          0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/R-kvfree_rc+
    7 root          0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/R-slub_flus+
    8 root          0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/R-netsns
   12 root         20   0      0      0      0 I   0.0   0.0   0:00.00 kworker/R-rcu_gp
```

Выбираем процесс 1 и меняем ему приоритет на -10

```
MiB Mem : 1968.7 total, 74.5 free, 978.1 used, 1101.3 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 990.6 avail Mem
Renice PID 1 to value -10
  PID USER      PR  NI  VIRT  RES  SHR S %CPU  %MEM    TIME+  COMMAND
 1951 vboxuser   20   0 3462708 382340 144156 S  13.0  19.0   0:21.00 gnome-s+
 3916 vboxuser   20   0  554060  53088  42316 S   4.0   2.6   0:00.82 gnome-t+
 3785 root         20   0      0      0      0 I   0.3   0.0   0:00.17 kworker+
```

В столбцах pr и ni можно увидеть изменения.

```
 1951 vboxuser   20   0 3454572 320716  82700 S   0.7  15.9   1:05.59 gnome-s+
  284 root         20   0      0      0      0 I   0.3   0.0   0:03.72 kworker+
 3916 vboxuser   20   0  624392  40424  28500 S   0.3   2.0   0:03.43 gnome-t+
 4179 root         20   0  14536   5828   3652 R   0.3   0.3   0:00.32 top
    1 root        10 -10  23272  14076   9476 S   0.0   0.7   0:33.07 systemd
    2 root         20   0      0      0      0 S   0.0   0.0   0:00.00 kthreadd
    3 root         20   0      0      0      0 S   0.0   0.0   0:00.00 pool_wo+
```

Проделываю то же самое с процессом 2.

```
MiB Mem : 1968.7 total, 526.9 free, 993.8 used, 632.6 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 974.9 avail Mem
Renice PID 2 to value -10
  PID USER      PR  NI  VIRT  RES  SHR S %CPU  %MEM    TIME+  COMMAND
 1951 vboxuser   20   0 3454572 320716  82700 S   0.7  15.9   1:06.52 gnome-s+
  284 root         20   0      0      0      0 I   0.3   0.0   0:03.90 kworker+
  835 root         20   0 1865340  39624  24080 S   0.3   2.0   0:46.98 snapd
 3899 root         20   0      0      0      0 I   0.3   0.0   0:01.25 kworker+
 4179 root         20   0  14536   5828   3652 R   0.3   0.3   0:00.49 top
    1 root        10 -10  23272  14076   9476 S   0.0   0.7   0:33.07 systemd
    2 root         20   0      0      0      0 S   0.0   0.0   0:00.00 kthreadd
    3 root         20   0      0      0      0 S   0.0   0.0   0:00.00 pool_wo+
    4 root          0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker+
    5 root          0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker+
    6 root          0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker+
    7 root          0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker+
    8 root          0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker+
 3810 root         20   0      0      0      0 I   0.3   0.0   0:00.24 kworker+
 4179 root         20   0  14536   5828   3652 R   0.5   0.3   0:00.51 top
    1 root        10 -10  23272  14076   9476 S   0.0   0.7   0:04.12 systemd
    2 root        10 -10      0      0      0 S   0.0   0.0   0:00.00 kthreadd
    3 root         20   0      0      0      0 S   0.0   0.0   0:00.00 pool_wo+
    4 root          0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker+
    5 root          0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker+
```

6. Получить список открытых файлов пользователя

```
vboxuser@ubuntu:~$ lsof -u vboxuser
```

```
/lib/x86_64-linux-gnu/libsharpvuv.so.0.0.1
gnome-she 1951 vboxuser mem REG      8,2    14344 1322820 /usr
/lib/x86_64-linux-gnu/libwayland-egl.so.1.22.0
gnome-she 1951 vboxuser mem REG      8,2    510592 1322808 /usr
/lib/x86_64-linux-gnu/libvulkan.so.1.3.275
gnome-she 1951 vboxuser mem REG      8,2    1189616 1322305 /usr
/lib/x86_64-linux-gnu/libepoxy.so.0.0.0
gnome-she 1951 vboxuser mem REG      8,2     22600 1322498 /usr
/lib/x86_64-linux-gnu/libkeyutils.so.1.10
gnome-she 1951 vboxuser mem REG      8,2     18504 1322241 /usr
/lib/x86_64-linux-gnu/libcom_err.so.2.1
gnome-she 1951 vboxuser mem REG      8,2     41804 1323961 /usr
/lib/x86_64-linux-gnu/girepository-1.0/Gsk-4.0.typelib
gnome-she 1951 vboxuser mem REG      8,2    570548 1323968 /usr
/lib/x86_64-linux-gnu/girepository-1.0/Gtk-4.0.typelib
gnome-she 1951 vboxuser mem REG      8,2     14972 1323975 /usr
/lib/x86_64-linux-gnu/girepository-1.0/NMA4-1.0.typelib
gnome-she 1951 vboxuser mem REG      8,2     19176 1313326 /usr
/lib/gnome-shell/Gvc-1.0.typelib
gnome-she 1951 vboxuser mem REG      8,2    340688 1323974 /usr
/lib/x86_64-linux-gnu/girepository-1.0/NM-1.0.typelib
gnome-she 1951 vboxuser mem REG      8,2     45580 1313328 /usr
/lib/gnome-shell/St-14.typelib
gnome-she 1951 vboxuser mem REG      8,2    212868 1324215 /usr
```

7. Получить текущее состояние системной памяти

```
vboxuser@ubuntu:~$ free
              total        used        free      shared  buff/cache   available
Mem:           2015944    1024332     485340        35696     696216     991612
Swap:              0              0              0
```

8. Получить справку об использовании дискового пространства.

```
vboxuser@ubuntu:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs           197M  1.7M  196M   1% /run
/dev/sda2       25G   5.6G   18G  24% /
tmpfs           985M    0  985M   0% /dev/shm
tmpfs           5.0M   8.0K   5.0M   1% /run/lock
tmpfs           197M  156K  197M   1% /run/user/1000
/dev/sr0        6.0G   6.0G    0 100% /media/vboxuser/Ubuntu 24.04.3 LTS amd64
```

9. Вывести информацию о каком-либо процессе, используя содержимое каталога /proc


```
vboxuser@ubuntu:~$ ls /proc/1
ls: cannot read symbolic link '/proc/1/cwd': Permission denied
ls: cannot read symbolic link '/proc/1/root': Permission denied
ls: cannot read symbolic link '/proc/1/exe': Permission denied
arch_status      fdinfo           ns               smaps_rollup
attr             gid_map          numa_maps       stack
autogroup        io              oom_adj         stat
auxv             ksm_merging_pages oom_score       statm
cgroup           ksm_stat        oom_score_adj   status
clear_refs       latency         pagemap         syscall
cmdline          limits          patch_state     task
comm             loginuid        personality      timers_offsets
coredump_filter  map_files       projid_map      timers
cpu_resctrl_groups maps            root            timerslack_ns
cpuset           mem            sched           uid_map
cwd              mountinfo       schedstat       wchan
environ          mounts          sessionid
exe              mountstats      setgroups
fd              net            smaps
vboxuser@ubuntu:~$
```

10. Вывести информацию о процессоре ПК, используя содержимое каталога /proc

```
vboxuser@ubuntu:~$ cat /proc/cpuinfo
processor       : 0
vendor_id      : GenuineIntel
cpu family     : 6
model          : 78
model name     : Intel(R) Core(TM) i7-6500U CPU @ 2.50GHz
stepping       : 3
microcode      : 0xffffffff
cpu MHz        : 2591.998
cache size     : 4096 KB
physical id    : 0
siblings       : 1
core id        : 0
cpu cores      : 1
apicid         : 0
initial apicid : 0
fpu            : yes
fpu_exception  : yes
cpuid level    : 22
wp             : yes
flags           : fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov
pat pse36 clflush mmx fxsr sse sse2 ht syscall nx rdtscp lm constant_tsc rep_goo
d nopl xtopology nonstop_tsc cpuid tsc_known_freq pni pclmulqdq ssse3 fma cx16 p
cid sse4_1 sse4_2 movbe popcnt aes xsave avx f16c rdrand hypervisor lahf_lm abm
3dnowprefetch pti fsgsbase bmi1 avx2 bmi2 invpcid rdseed adx clflushopt arat md_
clear flush_l1d arch_capabilities
bugs           : cpu_meltdown spectre_v1 spectre_v2 spec_store_bypass l1tf mds
swapgs itlb_multihit srbds mmio_stale_data retbleed gds bhi its
bogomips       : 5183.99
clflush size   : 64
```

11. Вывести список модулей, используемых в настоящий момент ядром ОС.

```
vboxuser@ubuntu:~$ lsmod
```

Module	Size	Used by
isofs	61440	1
snd_seq_dummy	12288	0
snd_hrtimer	12288	1
qrtr	53248	2
intel_rapl_msr	20480	0
intel_rapl_common	53248	1 intel_rapl_msr
intel_pmc_core	122880	0
pmt_telemetry	16384	1 intel_pmc_core
pmt_class	16384	1 pmt_telemetry
intel_vsec	20480	1 intel_pmc_core
snd_intel8x0	53248	1
snd_ac97_codec	196608	1 snd_intel8x0
ac97_bus	12288	1 snd_ac97_codec
snd_pcm	192512	2 snd_intel8x0,snd_ac97_codec
snd_seq_midi	24576	0
snd_seq_midi_event	16384	1 snd_seq_midi
polyval_clmulni	12288	0
snd_rawmidi	57344	1 snd_seq_midi
polyval_generic	12288	1 polyval_clmulni
ghash_clmulni_intel	16384	0
snd_seq	122880	9 snd_seq_midi,snd_seq_midi_event,snd_seq_dummy
sha256_ssse3	32768	0
sha1_ssse3	32768	0
aesni_intel	122880	0
snd_seq_device	16384	3 snd_seq,snd_seq_midi,snd_rawmidi
snd_timer	53248	3 snd_seq,snd_hrtimer,snd_pcm
joydev	32768	0
crypto_simd	16384	1 aesni_intel